

João Victor Mendes

Production Support Engineer (PSE) / SRE | Cloud · Observability · Database Reliability

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União da Vitória, PR, Brazil (UTC-3)

PROFESSIONAL SUMMARY

Production Support Engineer (PSE) and SRE supporting critical distributed platforms (FinTech and AI). Hands-on expertise in relational database saturation diagnostics (AWS Aurora MySQL, PostgreSQL), structural telemetry noise diagnosis, and critical production incident response with a focus on detection validation to reduce MTTD and root-cause post-mortems.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (Aurora MySQL/RDS, EKS, CloudWatch, SSM), Kubernetes (Karpenter, ArgoCD, Topology Spread Constraints), Docker, Linux (systemd, boot/network troubleshooting).
- **Observability & SRE:** New Relic (NRQL, NerdGraph API, baseline/alerting), OpenTelemetry, Incident Response (P1/P2), Post-Mortem / RCA (ITIL), Detection Engineering & Runbooks, SLA.
- **Databases:** AWS Aurora MySQL (Performance Insights, connection pooling, parameter groups), PostgreSQL (WAL, replication slots, bloat, EXPLAIN ANALYZE, pg_class).
- **IaC & Automation:** Terraform, Terragrunt, Python, Bash, PowerShell, SQL, Git.
- **Languages:** English (B2 Certified, BRASAS — daily technical documentation and incident response), Portuguese (Native).

PROFESSIONAL EXPERIENCE

Monest | Credit Recovery FinTech & AI Platform Aug 2026 – Present
Production Support Engineer (PSE) / SRE Remote, Brazil

Engineering reliability and observability for a distributed cloud platform powering AI agents, VoIP dialers, and microservices.

- **Database Saturation Diagnostics & Capacity (Aurora MySQL):** Contributed to diagnosing CPU and connection saturation on a production Aurora cluster, separating load concentrated in a few queries, triggered by a large batch, from connection-pool amplification across microservices (instances × pool size above the cluster ceiling); the analysis supported prioritizing RDS Proxy.
- **Detection Engineering & Alerting Governance (New Relic / Terraform):** Audited hundreds of alert conditions via GraphQL API and identified an unsegmented policy driving roughly 40% of incident volume and a baseline pattern that never fires on sparse workloads; authored Aurora monitoring policies in Terraform calibrated on 7 days of measurements and recalibrated load alarms that sat in breach more than a third of the time and suppressed new pages.
- **External Integration Root Cause (API / Observability):** Diagnosed intermittent failures in an external-service integration: schema validation inside the try block masked HTTP 200 responses as 500 errors; comparing logged errors against genuinely failed HTTP calls separated the hypotheses, and after the owning team's schema fix errors dropped from 9,428 to 3.
- **Data Pipeline Incident Response (Airflow / Docker / SSM):** Led post-mortem for a data pipeline outage caused by disk exhaustion; bypassed SSH blockers via SSM without admin privileges, validated with historical data that an NRQL deadman alert would have detected the failure about 1h40 before human detection, and fixed an IaC disk rule where cross-mount averaging masked ~99% root saturation.
- **Scheduler Outage & Host Hardening (Systemd / SSM / Linux):** Traced root cause of a prolonged job-scheduler outage: monitoring checked proxy availability, not execution; during a controlled reboot, diagnosed boot races and missing restart policies, extended resilient systemd units and configured Slack failure alerts via OnFailure, with credentials in SSM Parameter Store.
- **K8s Resiliency (EKS / Karpenter / ArgoCD):** Helped resolve an AI observability stack outage after Spot node termination; strengthened cross-node replica distribution via ArgoCD enforcing strict topologySpreadConstraints (DoNotSchedule), preventing Karpenter single-node bin-packing.

Pedroso Automação | B2B fiscal compliance ERP SaaS
Technical Support Engineer (L2/L3)

June 2024 – Aug 2026
União da Vitória, PR

Direct L2/L3 support for 500+ corporate environments (1,500+ end-users). Maintained full independence on solo rotations, resolving P1/P2 incidents without escalation through deep structural analysis.

- **Eliminated WAL retention risk across the fleet:** built scheduled PowerShell monitor for PostgreSQL detecting and dropping inactive orphan replication slots (>2GB or >30 min inactive); enforced hard server-side limit via `max_slot_wal_keep_size` in PG13+.
- **Automated disaster recovery and PostgreSQL provisioning:** parallelized cluster recovery across 30 databases (cutting downtime from hours to <5 min via `postmaster.pid` cleanup and WAL validation) and standardized Windows installation with runtime hardening (~2h down to <7 min).
- **Diagnosed 1GB/month disk bloat** via `pg_class` and `pg_toast`; isolated root cause to encrypted audit trail filepaths serialized across thousands of characters; reduced restore time by 60% (17 min down to 6:58) via `pg_restore -j 4`.
- **Built 70M-record ETL pipeline** (15GB+ government dataset) achieving a 60% processing time reduction; implemented automated corporate tax classification and XML invoice auditing with staging in PostgreSQL.
- **Protected \$100k USD (R\$500k) in critical inventory (zero data loss)** by reverse engineering an undocumented legacy database (Firebird); reprocessed 20 years of transaction logs to restore state with full consistency, delivered the same day the schema was first accessed.
- **Established reliability and mentoring culture:** authored 20+ video guides, runbooks, and ITIL post-mortems accelerating junior onboarding; engineered LLM-assisted triage with database context injection, reducing fiscal diagnostic time from 30+ to <5 min.

Girafa Comunicação Interativa | Digital Agency
Cloud Infrastructure Analyst & Web Operations

Sep 2022 – Apr 2024
Remote, Brazil

Operated Cloud infrastructure and web operations for 70+ corporate clients (AWS Route 53, Registro.br, Locaweb).

- **Cloud Operations & Incident Resolution:** Managed corporate DNS, WAF, and zones across AWS Route 53 and CloudFront for 70+ domains; served as international liaison in 15+ complex migrations with external registrars (GoDaddy, AWS, Namecheap) and isolated propagation/caching failures.
- **Security Incident Investigation:** Audited Git histories and access logs to identify security breaches; revoked compromised sessions and rotated credentials across 10+ corporate environments, preventing incident recurrence.

EDUCATION & PEDIGREE

FGV — Data Science & Economics (2020–2022)
Full-ride national OBMEP scholarship. Completed 2 years of rigorous coursework before transitioning to full-time production engineering.

Rio de Janeiro, Brazil

IFPR — Systems Analysis & Development (2023–2024)
Completed 3 semesters emphasizing computer science fundamentals alongside full-time production engineering work.

União da Vitória, Brazil

Silver Medalist — Brazilian Public School Mathematics Olympiad (OBMEP, 2016)
Top 0.01% of 18M+ participants nationally. Bronze Medalist (2014); 2x Honorable Mentions.

National